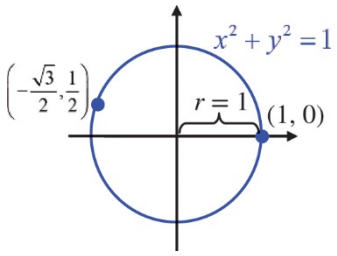
Use the Unit Circle to Define Trigonometric Functions

Unit Circle: The circle of radius 1 unit, centered at the origin.
Each point (x, y) on the unit circle satisfies the equation $x^2 + y^2 = 1$.



1. Show that the point $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ lies on the unit circle.

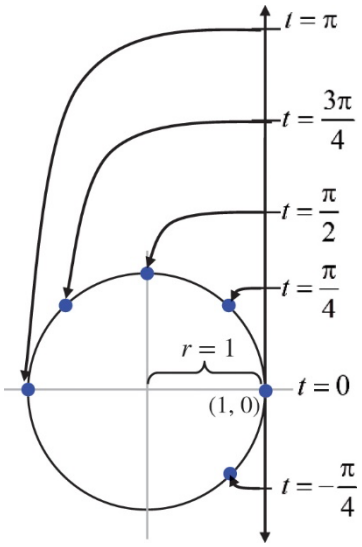
Suppose we wrap the real number line around the unit circle by placing the point $t = 0$ from the number line on the point (1, 0) on the circle.

Positive real numbers from the number line ($t > 0$) wrap onto the unit circle in the counterclockwise direction.

Negative numbers from the number line ($t < 0$) wrap onto the unit circle in the clockwise direction.

This associates each real number t with a unique point (x, y) on the unit circle.

This enables us to define trigonometric functions as *circular functions*.



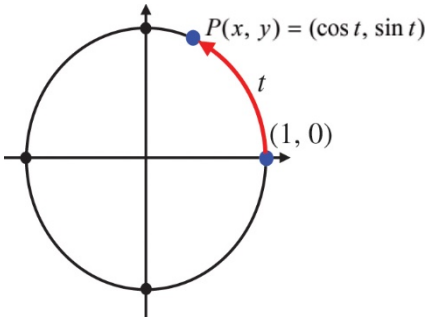
Unit Circle Definition of Trigonometric Functions

Let $P(x, y)$ be the point associated with a real number t measured along the circumference of the unit circle from the point (1, 0).

$\sin t = y$ $\quad \csc t = \frac{1}{y} \ (y \neq 0)$

$\cos t = x$ $\quad \sec t = \frac{1}{x} \ (x \neq 0)$

$\tan t = \frac{y}{x} \ (x \neq 0)$ $\quad \cot t = \frac{x}{y} \ (y \neq 0)$



If P is on the y-axis [either (0, 1) or (0, −1)], then $x = 0$ and the tangent and secant functions are undefined.

If P is on the x-axis [either (1, 0) or (−1, 0)], then $y = 0$ and the cotangent and cosecant functions are undefined.

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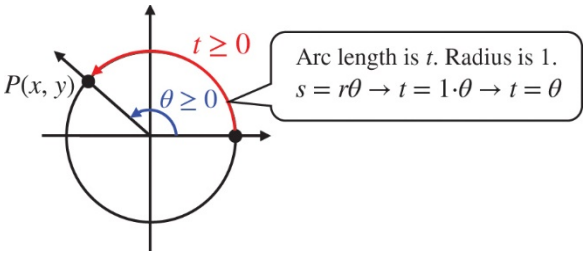
2. Suppose that the real number t corresponds to the point $P\left(\frac{5}{13}, -\frac{12}{13}\right)$ on the unit circle.

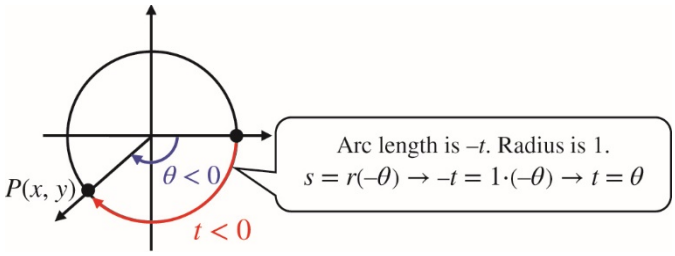
Evaluate $\sin t$, $\cos t$, and $\tan t$.

Evaluate Trigonometric Functions Using the Unit Circle

Let $P(x, y)$ be the point on the unit circle associated with a real number $t \geq 0$. Let $\theta \geq 0$ be the central angle in standard position measured in radians with terminal side through P and arc length t.

Because the radius of the unit circle is 1, the arc length formula $s = r\theta$ becomes $t = 1 \cdot \theta$ or simply $t = \theta$.





Trigonometric Functions of Real Numbers and Angles

If $\theta = t$ radians, then

$\sin t = \sin\theta$	$\cos t = \cos\theta$	$\tan t = \tan\theta$
$\csc t = \csc\theta$	$\sec t = \sec\theta$	$\cot t = \cot\theta$

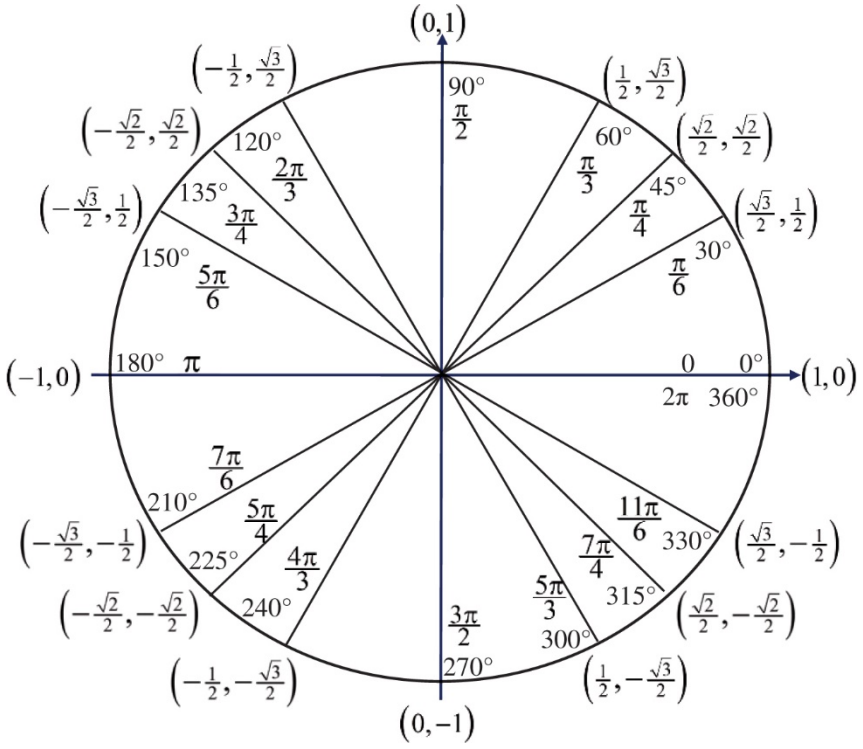
The trigonometric functions of an angle θ can now be evaluated using the unit circle.

Angle θ can be measured in radians or degrees because the values of the functions depend on the *coordinates* of $P(x, y)$ where the terminal side of θ intercepts the unit circle.

As a result, the angle may be measured in radians or degrees.

Note the symmetry of the unit circle.

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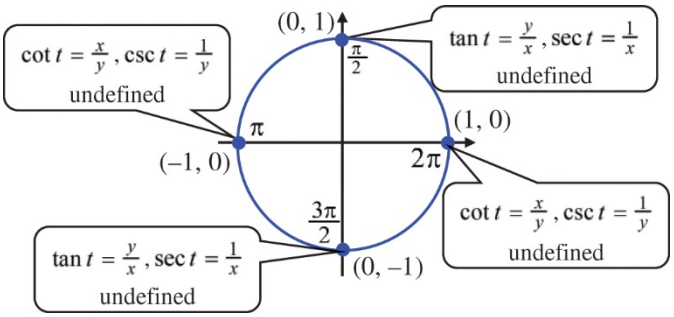
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3. Identify the ordered pair on the unit circle that corresponds to the number $t = \frac{5\pi}{3}$. Then evaluate sin t and sec t.

4. Evaluate cos t and tan t for $t = -\frac{5\pi}{6}$.

Identify the Domains of the Trigonometric Functions

$\sin t = y$ and $\cos t = x$ have no restrictions on their domain.



$\tan t = \frac{y}{x}$ and $\sec t = \frac{1}{x}$ are undefined for all values of t corresponding to the points (0, 1) and (0, −1) on the unit circle.

$\cot t = \frac{x}{y}$ and $\csc t = \frac{1}{y}$ are undefined for all values of t corresponding to the points (1, 0) and (−1, 0) on the unit circle.

Function	Domain	Notes
$f(t) = \sin t$, $f(t) = \cos t$	All real numbers	No restrictions
$f(t) = \tan t$, $f(t) = \sec t$	$\left\{ t \mid t \neq \frac{(2n+1)\pi}{2} \text{ for all integers } n \right\}$	Exclude odd multiples of $\frac{\pi}{2}$ (90°).
$f(t) = \cot t$, $f(t) = \csc t$	$\{ t \mid t \neq n\pi \text{ for all integers } n \}$	Exclude multiples of π (180°).

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Apply the Period and Properties of Trigonometric Functions

Periodic Function: A function f is **periodic** if $f(t+p)=f(t)$ for some constant p.
(f repeats at regular intervals)
The smallest positive value p for which f is periodic is called the **period** of f.

Sine, cosine, cosecant, and secant are periodic with period 2π.
Tangent and cotangent are periodic with period π.

Periodic Properties of Trigonometric Functions			
Function	**Period**	**Property**	
Sine	2π (360°)	$\sin(t+2\pi)=\sin t$	$\sin(\theta+360°)=\sin\theta$
Cosine	2π (360°)	$\cos(t+2\pi)=\cos t$	$\cos(\theta+360°)=\cos\theta$
Cosecant	2π (360°)	$\csc(t+2\pi)=\csc t$	$\csc(\theta+360°)=\csc\theta$
Secant	2π (360°)	$\sec(t+2\pi)=\sec t$	$\sec(\theta+360°)=\sec\theta$
Tangent	π (180°)	$\tan(t+\pi)=\tan t$	$\tan(\theta+180°)=\tan\theta$
Cotangent	π (180°)	$\cot(t+\pi)=\cot t$	$\cot(\theta+180°)=\cot\theta$

Use the unit circle and the period of the function to evaluate the expression if possible.

5. $\cos(-315°)$

6. $\tan\left(-\frac{\pi}{2}\right)$

7. $\sin\frac{17\pi}{6}$

8. $\sec 510°$

9. $\cot\frac{7\pi}{6}$

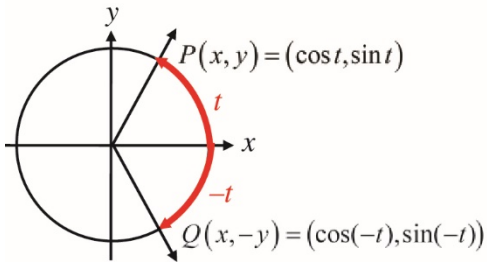
10. $\csc 540°$

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A function f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$.

$\cos(-t) = \cos t = x$

The cosine function is an even function.



$\sin t = y$ but $\sin(-t) = -y$. Thus, $\sin(-t) = -\sin t$.

The sine function is an odd function.

$\tan(-t) = \dfrac{-y}{x}$ and $\tan t = \dfrac{y}{x}$. Therefore $\tan(-t) = -\tan t$.

The tangent function is also an odd function.

Summary: cos *t* and sec *t* are even functions. sin *t*, csc *t*, tan *t*, and cot *t* are odd functions.

Use the properties of the trigonometric functions to simplify.

11. $\tan(t+\pi)\cdot\cos(t+2\pi)$

12. $3\sin t + \sin(2\pi - t)$

13. $\sin(-t)\cdot\sec t - \tan(-t)$

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